

Homework A3 | ECON 0100 | Fall 2026

Due: Sunday, Sep. 6

Homework is designed to both test your knowledge and challenge you to apply familiar concepts in new applications. Answer clearly and completely, and show your work so you can later understand your thought process. You are welcomed and encouraged to work in groups as long as your work is your own. Submit your answers on Gradescope when you're finished: each answer there is a selection.

The Hogsmeade Candy Shop Saga

Honeydukes and Zonkos are two candy shops in Hogsmeade. Both shops make two types of popular candy, Chocolate Frogs (F) and Ice Mice (M). Devoting all their resources to one good, Honeydukes can make 100 pounds of F or 75 pounds of M each week, and Zonkos can make 200 pounds of F or 100 pounds of M .

Use the production table and opportunity cost table you built in Homework A2.

Q1 | Specialization

- a) If the two shops want to jointly produce more, which good should each shop specialize in?
- b) If each shop fully specializes, how many pounds of each candy do the two shops make in total each week?

Q2 | Terms of Trade

The two shops decide to specialize, trade, and sell each other's goods at their shops.

- a) What's an exchange rate that would facilitate a trade of Ice Mice for Chocolate Frogs? Give it in pounds of F per pound of M .

b) Explain why any exchange rate between the two shops' opportunity costs of Ice Mice would work.

Q3 | Accept or Reject

Suppose both shops specialize according to their comparative advantage: Zonkos makes only Chocolate Frogs and Honeydukes only Ice Mice.

- a) Zonkos offers 1F for each 1M. Which shop rejects the offer, and why?
- b) Honeydukes counters with 1M for 3F. Which shop rejects the counteroffer, and why?
- c) They settle on 1M for $3/2$ F and trade 40M for 60F. How much of each candy does each shop end up with?
- d) Use a graph or algebra to show whether each shop ends up beyond its own PPF.

Q4 | Changing Labor

Zonkos' owner decides to go half-time, cutting Zonkos' labor in half.

- a) Set up Zonkos' old and new PPF on the same graph. What are Zonkos' new maximum weekly outputs, and what is its new opportunity cost of 1F?
- b) Write a short description of how this change affects the trade with Honeydukes.

<!-- The questions are free response on paper; students submit selections on Gradescope. The selection form and answer key live in Homework_A3_gradescope.md. ANSWER KEY (not for students) OC table: Honeydukes $OC(1F) = 3/4$ M, $OC(1M) = 4/3$ F; Zonkos $OC(1F) = 1/2$ M, $OC(1M) = 2$ F. Q1a: Zonkos in Chocolate Frogs, Honeydukes in Ice Mice Q1b: 200F and 75M Q2a: any rate strictly between $4/3$ and 2 F per M (e.g. $3/2$ F

per M) Q2b: Honeydukes accepts anything above $\frac{4}{3}$ F per M (its own cost of 1M); Zonkos accepts anything below 2 F per M (its own cost of 1M); between them both do better than their self-trade Q3a: Honeydukes rejects: 1F per M is worse than the $\frac{4}{3}$ F it can turn 1M into itself Q3b: Zonkos rejects: 3F per M is worse than the 2F it costs Zonkos to make 1M itself Q3c: Honeydukes 60F and 35M; Zonkos 140F and 40M Q3d: both beyond: Honeydukes' PPF gives at most 30M alongside 60F (75 - 45); Zonkos' gives at most 30M alongside 140F (100 - 70) Q4a: 100F or 50M; OC of 1F still $\frac{1}{2}$ M (parallel shift in) Q4b: PPF shifts in without changing slope, so opportunity costs and the acceptable range of exchange rates are unchanged; but Zonkos can now trade away at most 100F, so the volume of trade falls $\rightarrow$